

Macarthur Diby

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SUMMARY

Software engineer with a B.S. in Computer Science (3.7 GPA) from Texas Tech, currently at Dell Technologies. Builds full-stack and AI/ML systems: LLM distillation pipelines, real-time computer vision, and multi-service cloud deployments. Open to relocation. Fluent in English and French.

PROFESSIONAL EXPERIENCE

- Dell Technologies** · *Senior Technician, Technical Support Engineer* · Austin, TX Jun 2025 – Present
- Earned Top Performer Gold in two months, ranking in the top tier out of 5,000+ agents company-wide.
 - Contributed to AI-driven diagnostic tooling alongside software engineers, supporting testing and implementation cycles that reduced case resolution time by 20%.
 - Maintained 98%+ customer satisfaction rate diagnosing complex enterprise hardware and software failures.
- WoundCentrics** · *Software Engineer* · Lubbock, TX Jan 2025 – May 2025
- Built Python automation tools for billing workflows and clickstream tracking, cutting processing time 30% and manual work 40%.
 - Integrated third-party healthcare APIs and built real-time claims dashboards, reducing rejection rates by 25%.
- IT Help Central, Texas Tech University** · *IT Systems Technician* · Lubbock, TX Jul 2022 – Dec 2024
- Resolved 300+ ServiceNow tickets for 150+ students; trained incoming technicians; repeat incidents dropped 15% and resolution time improved 25%.

PROJECTS

- LLM Knowledge Distillation Pipeline** [\[GitHub\]](#) | *Python, PyTorch, HuggingFace Transformers, Anthropic API, Streamlit*
- Used Claude Haiku to label 1,160 support tickets with soft probability distributions, then fine-tuned DistilBERT as a student model using KL divergence + cross-entropy distillation loss ($T=4$).
 - Student runs 65x faster than the teacher at near-zero serving cost, retaining 87% of accuracy across 8 support categories.
- Driver Drowsiness Detection System** [\[GitHub\]](#) | *Python, OpenCV, dlib, Flask, React Native (Expo), Supabase*
- Real-time drowsiness detection using Eye Aspect Ratio (EAR); three-component architecture: dlib/OpenCV tracker, Flask REST API, and React Native app, with live session updates via Supabase Realtime.
 - Three-tier alert system (NORMAL / WARNING / DANGER) driven by event frequency over rolling time windows; no polling on the client side.
- AI Debate System** [\[GitHub\]](#) | *Node.js, React, OpenAI / Gemini / Groq APIs, AWS EC2 & S3*
- Multi-round debate platform where six AI models argue user-defined topics; separate AI referee models score each round, with the backend on AWS EC2 and frontend on S3.
- Patient Appointment System** [\[GitHub\]](#) | *Python, Flask, SQLite, JavaScript, HTML/CSS*
- Full-stack scheduling app with user authentication, department routing, and role-based access, built and deployed solo.
- Elevator Scheduler** [\[GitHub\]](#) | *C++, multithreading, mutexes, condition variables*
- Concurrent elevator dispatcher with thread-safe request queues using POSIX mutexes and condition variables; hybrid algorithm cut average wait time 20–30% vs. distance-only baseline.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Flask, React Native, PyTorch, OpenCV, Scikit-learn, HuggingFace Transformers, Streamlit

Cloud & DevOps: AWS (EC2, S3), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Linux

Databases & Tools: MySQL, PostgreSQL, MongoDB, Supabase | Git, REST APIs, OpenAI API, Gemini API, Anthropic API

EDUCATION

B.S. Computer Science, Minor in Mathematics · Texas Tech University, Lubbock, TX Dec 2024

GPA: 3.7 | 5x Dean's List | Presidential List | Achiever Scholarship

ACTIVITIES & VOLUNTEER WORK

- NSBE, Texas Tech Chapter Recruiter | Child's Heart, Côte d'Ivoire · Youth volunteer, 50+ children served